

303Modbus13 RTU Control

Python 3.10+
Modbus RTU
GitHub flexTech-KRK/303Modbus13
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Desktop GUI application for controlling and diagnosing the **Ideaflex 303Modbus13** relay module over **Modbus RTU** (serial/COM).

Git repository: github.com/flexTech-KRK/303Modbus13

Designed for customers and integrators — quick module bring-up without a TCP/IP gateway.

Features

Feature	Description
2 relays	CH1–CH2 ON/OFF, bulk Both ON / Both OFF
2 opto inputs	Live status of IN1–IN2 (FC 0x02)
T/H sensor	Temperature and humidity readout (if installed)
Auto-refresh	Cyclic status polling every 1 s
Address change	Slave ID configuration via Modbus broadcast
Modbus console	Full access to FC 01–06, 0F, 10
Documentation	Register map: docs/REGISTERS.md · on GitHub

Requirements

- Windows 10/11 (or Linux with a serial port)
 - Python 3.10 or newer
 - USB ↔ RS-485 converter (or built-in COM port)
 - 303Modbus13 module on the RS-485 bus
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Quick start

1. Clone the repository

```
git clone https://github.com/flexTech-KRK/303Modbus13.git cd 303Modbus13
```

Online sources: [README.md](#) · [REGISTERS.md](#)

2. Virtual environment

```
python -m venv .venv # Windows .venv\Scripts\activate # Linux/macOS source .venv/bin/activate
```

3. Install dependencies

```
pip install -r requirements.txt
```

4. Run the application

```
python app.py
```

5. Connect to the module

1. Connect the module to a COM port (e.g. COM4 via USB-RS485).
2. Select the COM port and click **Connect**.
3. Default Slave ID: FF (255, factory address).
4. Open the **Control** tab and test the relays.

Communication parameters

Parameter	Value
Protocol	Modbus RTU
Baud rate	9600
Format	8N1
Slave ID (relays)	0xFF (255) factory default, then e.g. 0x01
Slave ID (T/H sensor)	0x02
Timeout	3.0 s (serial port / pymodbus)
Retries	3
Post-write coil delay	0.35 s

At 9600 baud with a USB-RS485 converter, **do not use a 0.5–1.0 s timeout** — the module responds correctly but needs more time. Details, frames, and troubleshooting: [docs/REGISTERS.md](#).

Opto inputs (IN / COM)

Inputs are **opto-isolated**. Both connections are required:

- **COM** → GND (0 V) of the signal source
- **INx** → signal voltage (+12...24 V DC relative to COM)

Input **COM** is **not** the same as the board power GND — it is a separate isolated circuit. Voltage on a relay contact alone, without wiring **INx** and **COM** on the module, will not produce `True` in Modbus (FC 0x02).

Command-line example (pymodbus)

Control relay CH1:

```
from pymodbus.client import ModbusSerialClient client = ModbusSerialClient( port="COM4",
baudrate=9600, parity="N", stopbits=1, bytesize=8, timeout=3.0, retries=3, ) client.connect()
# Turn CH1 ON (new module: slave=255 / 0xFF) client.write_coil(address=0, value=True,
slave=255) # Read opto inputs (count=8 in frame; result: IN1, IN2, ...) inputs =
client.read_discrete_inputs(address=0, count=8, slave=255) print(inputs.bits[:2])
client.close()
```

Project structure

Path	Description
<code>app.py</code>	Tkinter GUI — application entry point
<code>modbus_device.py</code>	Modbus RTU communication layer
<code>requirements.txt</code>	Python dependencies
<code>README.md</code>	This file
<code>scripts/generate_pdfs.py</code>	Build PDF documentation
<code>docs/REGISTERS.md</code>	Full Modbus register map
<code>docs/REGISTERS.pdf</code>	PDF version of register map
<code>docs/README.pdf</code>	PDF version of this file

Generate PDFs

```
pip install markdown xhtml2pdf python scripts/generate_pdfs.py
```

Output: docs/README.pdf and docs/REGISTERS.pdf.

First-time module setup

A new module ships with Slave ID **0xFF (255)**. To set address **0x01**:

1. Connect with Slave ID **FF**.
2. On the **Connection** tab, enter new address **01** and click **Save new address**.
3. Disconnect and reconnect with Slave ID **01**.

Alternatively — via the advanced console:

```
Function: Write Multiple Registers (0x10) Address: 0 Values: 1 Slave: 0
```

Modbus TCP/RTU gateway

The module can also be used through a network gateway (e.g. USR-M0) with Modbus TCP. The register map is identical — only the transport layer changes (IP instead of COM).

TCP/RTU application version: local repo `ModBus_ReLay_303Modbus13` (RTU/COM version: [303Modbus13 on GitHub](#)).

Troubleshooting

Symptom	Solution
Cannot open COM port	Check Device Manager → COM ports; close other apps using the port
No response from device	Check RS-485 (A/B), Slave ID (FF or 01), timeout 3 s
Opto inputs always off	Wire COM → GND and INx → +signal (isolated circuit — see REGISTERS.md)
T/H sensor shows not connected	Sensor is optional — relay module works without it
Address change error	Use Slave=0 (broadcast)

License

© **Ideaflex sp. z o.o.** — software provided to customers with the 303Modbus13 module.

Contact

If you have any problems or questions, contact **FlexTech_KRK** on [Allegro](#) or email biuro@ideaflex.pl.